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Dear Editor,

I wish to resubmit the manuscript *Plants in limbo – the theory of detrital photosynthesis* for consideration as a Theories article in *Quantitative Plant Biology*. Based on the received constructive criticism, I have changed the graphical abstract, designed a fourth figure to illustrate the proposed theory, rewritten large parts of the manuscript, and cited additional studies. Data and code remain available at github.com/lukaseamus/plant-limbo. I have addressed the reviewers' comments in detail in italics below. Please note that a mathematical model of decomposition building on this theory has now been published in *Ecological Modelling*: doi.org/10.1016/j.ecolmodel.2026.111549. I have cited this model paper in the manuscript under consideration, and the model paper cites the manuscript as "in review" with *Quantitative Plant Biology*. I encourage the editor and reviewers to consult the paper for further context on the importance of this manuscript.

Reviewer: 1

Comments to the Author

Peer Review of Manuscript "Plants in limbo – the theory of detrital photosynthesis"
Quantitative Plant Biology for *Quantitative Plant Biology*

Summary

The author of this paper conducted a literature review and meta-analysis of data on the persistence of photosynthesis in excised/dislodged tissues from a variety of plants and seaweeds. They also conducted a short lab experiment testing the photosynthetic decay of seagrass tissues. The author then determined, relative to initial baseline, the photosynthetic half-life of the tissues included in the literature review. The author goes on to state that seaweeds are unique in their ability to remain photosynthetic for long periods following excision, compared to other plants.

Main comments

- I am struggling with the framing of this paper- the fact that it's submitted to a plant focused journal and the title indicates to me that the paper would focus on detrital photosynthesis more broadly. So I was surprised to see that it does mainly focus on kelp/blue carbon, particularly in the abstract and discussion. Broadening/shifting the framing in would help the paper appeal to a wider audience, I think that there are plant scientists who would be interested in this paper. However I know that the author's expertise is in kelp, I would just consider if publishing in a marine journal would be a better fit then. I would personally change the title to reference macroalgae/kelp/blue carbon if the framing is kept the same.

I thank the reviewer for their thoughts. I have edited the abstract to exclude “blue carbon” (now only in the keywords) and reworded the introduction and discussion to make no mention of blue carbon and reduce the focus on macroalgae. In principle I am happy to further broaden the scope of the introduction and conclusion while retaining the general storyline. However, it is unclear to me what exact framing the reviewer is after and what information they believe is currently missing. I kindly ask the reviewer or editor to explain what additional details of interest to plant scientists more broadly they would like included and I will of course oblige.

I maintain that this paper has a broad botanical scope. Kelps are merely the first study organisms that provided evidence for detrital photosynthesis and thus provide the rationale for this study. Far from focussing on kelps, the results include data on various terrestrial plants as well as freshwater and marine Alismatales in addition to macroalgae (all of which happen to be seaweeds). In fact, the literature review was not restricted to any particular plant group and the majority of the 92 studied species listed in Table S1 are terrestrial plants.

The reason the discussion necessarily focusses on macroalgae is because the study highlighted them as exceptional compared to vascular plants. This uniqueness has far-reaching implications for aquatic biogeochemistry of which blue carbon is an important subject area. A marine or even aquatic journal would not be suitable because most of the plants discussed here are not aquatic. I therefore also struggle to see how including a term as specific as “kelp” or “seaweed” in the title could benefit the story. Rather, I believe this would misrepresent the study.

- I think that the introduction can be improved upon by adding clarity and specificity to the author’s writing. The “bones” are there, but I find some sentences/lines of thought hard to follow. See line by line comments.

Please see line-by-line comments addressed below.

- I would also like to see the author directly indicate the knowledge gap that this study is filling in the introduction, have there been any other papers specifically looking at detrital photosynthesis/viability? The author references that this topic is referred to under different names- what are the different names? Are there similar papers published on this topic under different names?

I kindly refer the reviewer to the expanded last paragraph of the introduction where the aims are laid out (ll. 105ff.). The knowledge gaps are indicated throughout the introduction (e.g. “There is currently no reference point for detrital photosynthesis” in l. 56 and “seagrass detrital photosynthesis remains to be directly investigated” in ll. 102f.). All key existing phycological studies on detrital photosynthesis are cited along with some of the numerous post-excision physiology examples from the plant physiology literature. I clarified that this article is the first attempt to holistically study detrital photosynthesis (l. 107).

- o The author found a large number of papers which measured photosynthesis in “dead” materials, what is the motivation of these studies?

I am not entirely sure which papers the reviewer is referring to. None of the studies measuring photosynthesis assumed the tissue to be “dead” a priori or else they would have not conducted the experiments. The rationale for the 129 reviewed studies varies. I have clarified this in ll. 67f. Most studies on terrestrial and freshwater plants are interested in the influence of senescence on various

aspects of plant physiology. Senescence is mimicked by excision. For most studies on seaweeds and seagrasses, measurement of physiology is tangential to the main aim of measuring decomposition. Only recently has the explicit aim of some studies been to understand detrital photosynthesis (e.g. doi.org/10.1007/s00227-023-04289-y, doi.org/10.1093/aob/mcad167).

- Please include the methods section for the seagrass degradation experiment in the main text.

I had previously included the seagrass methods in the main body, but this distracted from the holistic nature of the story and exceeded the word limit set by Quantitative Plant Biology. Since the seagrass experiment only contributes 81 observations (<1% of all observations in the meta-analysis) but the seagrass methods are 1951 words long, I moved them to the supplement where they are still accessible to the interested reader and don't distract from the story. I will of course oblige and move the seagrass methods back to the main body at the discretion of the editor who must weigh up their importance against the journal's word count limit.

- I have some questions and concerns about the statistical methods, discussed in line by line comments below.

Please see line-by-line comments addressed below.

o In particular, I do not agree "Detrital photosynthesis cannot conceivably exceed baseline photosynthesis under identical conditions", if an entire macroalgae plant gets dislodged from the substrate and floats to the surface it would experience more optimal light conditions and could photosynthesize more effectively, no?

This is exactly why I specified "under identical conditions" (l. 135). As the reviewer notes, with a change in conditions, such as increased light, photosynthesis may increase. I have clarified this point. What was "An ill-defined baseline or changes in light, temperature, carbon, nutrients etc. may naturally lead to brief increases after detachment" now reads "An ill-defined baseline or changes in environment (e.g. when detritus floats and is exposed to more light) may naturally lead to brief increases after detachment" (ll. 135ff.).

- The author could do a better job of indicating the sample sizes and standard errors of means presented throughout the results, as this is a review it's important to know whether means/medians come from just a handful of datapoints or not.

Species- and group-specific sample sizes (n) are given in the captions of Figures 1, 2 and 3. I have now also added these to the caption of Table 1 and the body of Table 2 and refer to sample sizes in Table 2 in the caption of Table 3. Means and standard deviations reported in text are parameter estimates derived from 80 thousand Markov chain Monte Carlo (MCMC) samples (see Methods section, Meta-analysis subsection), not summary statistics based on a specific number of empirical observations. In multilevel models such as the one presented in Equation 2, information is shared between categories (e.g. plant groups, species and experiments). Inference therefore depends on all observations and a parameter does not have a specific sample size. There are consequently no sample mean and sample standard deviation. Often the number of group levels and hence parameters is more meaningful. I believe what to report must be decided on a case-by-case basis, which is why in the caption of Figure 3 I provide the number of experiments (corresponding to lines)

for photosynthetic half-life but the number of decay estimates (corresponding to points) for detrital half-life.

- The discussion section needs to be edited. In particular the author states that macroalgal physiology is the main driver of decay when that is most definitely not the case in situ. It could be one of the main drivers but microbial activity and grazing most definitely have an important impact on degradation as well, plus a suite of abiotic factors (that likely influence degradation by impacting microbial functioning). Plus physiology is not just photosynthesis, better to stay specific and replace these instances of “physiology” with “photosynthesis” throughout the manuscript.

I have substantially rewritten the discussion. I believe there to be a misunderstanding of the causal sequence and have attempted to additionally clarify this by designing a discussion figure with two directed acyclic graphs (Figure 4). The theory I am putting forward suggests that physiology uniquely determines macroalgal resistance to decomposition. As the reviewer points out, consumption by decomposers such as macrofauna (detritivores) and microbes (saprotrophs) is indeed a proximate cause of mass loss. I am suggesting that detrital photosynthesis can influence detrital mass directly through growth and repair and indirectly through induced defence against decomposers. The suggested causal paths are illustrated in Figure 4. I use the term “physiology” on purpose because it also encompasses induced defence and wound repair or regrowth which are driven by energy from detrital photosynthesis but may happen in the absence of photosynthesis. While abiotic factors such as light and temperature may indeed directly affect decomposers, I am suggesting that they mostly act via physiological causal path in the case of macroalgae.

- Generally, I do not agree that a theory is being presented here. A review and analysis of compiled data was done, but I think framing it as a theory goes too far. The fact that macroalgae can photosynthesize after dislodgement is not really a theory nor is this paper the first evidence of this. Lastly, I wonder if the author believes that we know enough about detrital photosynthesis, or are there still important unknowns- the fact that no red algae have been tested and very few seagrasses indicate to me that we need more data to inform our understanding.

I thank the reviewer for their perspective on this question of epistemology. I kindly refer them to the Quantitative Plant Biology article types section: cambridge.org/core/journals/quantitative-plant-biology/information/author-instructions/preparing-your-materials-article-types. Here theories are defined as articles that “focus on new concepts”, “build on meta-analysis”, “extract overarching trends”, “question existing dogma” and “derive novel conceptual models that make quantitative predictions”. More generally, scientific theory provides the conceptual framework for understanding empirical results on any level. The reviewer’s statement “framing it as a theory goes too far” suggests that they understand theory to mean grand theory (e.g. Darwin’s theory of evolution). What I am presenting is not grand theory but nonetheless theory. I recommend reading “What is theory?” in Cumming (2011, doi.org/10.1007/978-94-007-0307-0), Wacker (2008, doi.org/10.1111/j.1745-493X.2008.00062.x) and Maxwell (2023, doi.org/10.1177/17449871231178926) on this subject in various disciplines. Put simply, theory is any causal model “that explains previous findings and drives further empirical work” (Cumming 2011). According to all of these definitions, what I am presenting is a novel theory.

I do not believe that we know enough about detrital photosynthesis in specific plant groups such as seagrasses or green and red algae, nor do I say so anywhere in the manuscript. This is why I provide an outlook in the conclusion, to which I have now added “especially in underrepresented plants such

as seagrasses and seaweeds other than kelps" (ll. 576f.). I believe knowledge on specific groups is not a prerequisite for putting forward a theory. Rather, the theory is the prerequisite, providing the foundation for and guiding further empirical study into detrital photosynthesis.

Line by line comments

Abstract

See my comment above re: considering framing your article towards a broader plant/photosynthesis audience. Up to the author however the abstract is heavily focused towards kelp currently. The phenomenon of detrital photosynthesis is interesting and has important implications beyond just kelp/blue carbon.

Kelp is only mentioned once at the beginning to introduce the topic. I have removed "blue carbon" to the keywords (l. 25).

Specify that photosynthetic half-life values are means (or medians?)

I have specified that the reported values are medians (l. 17).

I would maybe chose one question/one half of the graphical abstract to present, for simplicity.

I have changed the graphical abstract based on the other reviewer's comments but retain both questions because I believe they reflect the two parts to this story. Whether or not decomposition hinges on physiology does not just depend on the persistence of photosynthesis in the detrital phase but also on the decomposition timescale.

Introduction

I find the first paragraph very hard to follow. I think best to restructure and remember that one paragraph=one idea. I would bring sentence starting L43 up.

I would like to point out that each of my four introductory paragraphs start with a topic sentence:

- 1. Seaweed detrital dynamics are influenced by detrital photosynthesis.*
- 2. There is currently no reference point for detrital photosynthesis.*
- 3. Seagrasses are suitable candidates for further investigation of detrital photosynthesis.*
- 4. Here I aim to formalise the theory of detrital photosynthesis.*

These sentences, often statements, are then elaborated in the paragraph. I am therefore already following the reviewer's suggestion of "one paragraph = one idea". The narrative naturally starts with seaweeds for which detrital photosynthesis was first reported (paragraph 1), raises the lack of a holistic investigation of detrital photosynthesis (paragraph 2), proposes that seagrasses could provide insight (paragraph 3) and lays out the resulting aims of the study (paragraph 4).

It seems that the aim of the sentence move suggested by the reviewer is to make earlier mention of "detrital photosynthesis". I have therefore changed the first topic sentence to "Seaweed detrital dynamics are influenced by detrital photosynthesis" rather than "physiology" and moved up the sentence starting "Detrital photosynthesis – defined as..." (ll. 38f.). The reason the reviewer may perceive the first paragraph to be disjointed is because it has to define the semantics of "detritus". This was requested by a previous reviewer who did not understand that detritus is not automatically dead. I have tried to improve the flow by moving the point on semantics to the end of the paragraph (ll. 52ff.).

L35: is this always true?

Yes, studies that have measured photosynthesis and sinking in floating seaweed have found this to be the case. I have cited these studies. I am not aware of any study that has provided evidence for dead floating seaweed.

L42: This sentence "I propose to continue using this term *sensu lato* for any detached plant tissue and *detritus sensu stricto* for detached and dead plant tissue." Feels very out of place, I don't think such a declaration should be made so early on, I'd suggest moving this to the discussion. Also please define *sensu lato* and *sensu stricto*. Also- why not just use a more broad term like "organic matter/materials" without implying either way dead/alive?

I included this definition because it was requested by a previous reviewer who did not understand why I used the term "detritus" to refer to detached plant material that is still alive. Since it is a definition that the remainder of the paper hinges on, the only suitable place is the first paragraph. Because the sentence in question purely concerns semantics and does not derive from an interpretation of my results, I do not believe the discussion would be an appropriate place for it. I have reworded to "I therefore continue to use the term..." (ll. 52f.) to clarify that this is definition and not interpretation and moved the sentence to the end of the first paragraph.

*I have removed *sensu lato* which is perhaps less commonly used. *Sensu stricto* is simply Latin for "in the strict sense" and since *sensu* is widely used in the scientific literature to abbreviate "in the sense of" and *stricto* practically translates itself into English, I think a secondary definition inside the definition would only add confusion.*

To me "detritus" is a broad term, effectively just a less technical way of saying "particulate organic matter". But to anyone who follows the Moore definition I cited it must also be dead, which is precisely what I am challenging here. There is no reason we should assume "detritus" means "dead", in the same way we should not assume "litter" or "particulate organic matter" to mean "dead".

L45: Fundamental impact of what? How does one distinguish between dead and living tissues anyway? This sentence confuses me.

I have reworded to "Its fundamental impact..." (ll. 40f.) referring to detrital photosynthesis in the previous sentences. The cited studies either use senescent tissues and compared them to fresh ones or experimentally removed physiology.

L47: Impaired? how so?

I specified "absent or impaired" (l. 42) but am not sure what else I can add here. The sentence compares live with dead or senescent tissue. Dead or senescent tissue by definition has absent or impaired physiology.

L49: What is likely due to growth + chemical defence? I believe this thought needs to be expanded upon for clarity.

I have reworded to "The effect of detrital photosynthesis on decomposition..." (l. 44), referring to the contrast laid out in the previous sentence.

L53: What do you mean by baseline? What would such a baseline look like?

I have reworded "baseline" as "reference point" (l. 56) to clarify that I am referring to observations of detrital photosynthesis in other plants against which to evaluate the observations on seaweeds. This sentence is a topic statement that is elaborated on and supported throughout the second paragraph, so it is hard to be more descriptive.

L57: ... for the purposes of studying degradation dynamics

I added "to measure decomposition" (l. 60).

L59: This is a lot of papers! What do they focus on? Do any of them discuss detrital photosynthesis at all? Would be good to indicate specifically how this review fills a knowledge gap not already filled by these review papers.

Have any of the empirical papers you included in your review directly studied detrital photosynthesis per se? Under a different name?

I have already limited the number of papers cited here to those vascular plant papers I cite elsewhere in the manuscript. Hence, I use "e.g." to indicate that there are many more (as can be seen in the supplementary references) and justify calling the amount of prior research "enormous". None of the cited papers are review papers. I am unsure what suggested this. As mentioned in text, these papers do not use the term "detrital photosynthesis". They do not use a specific term. They measure photosynthesis in detached/excised leaves for various reasons, but primarily to understand the senescence process. I have added a sentence explaining this (ll. 67ff.).

L64: what research trajectories?

Reworded as "...align these seemingly incongruous data in a formal comparison..." (ll. 69f.).

L64: ... formal comparison across photosynthetic organisms

Added "across plant groups" (l. 70).

L68: What do you mean by reference points? Also what does it mean to independently photosynthesize?

Changed "reference points" to "lack of knowledge on other plants" (ll. 73f.). I do not know how to express "ability ... to independently photosynthesise" more clearly. It simply means photosynthesis independent of other tissues, e.g. lamina independent of stipe and holdfast, leaf independent of root etc.

L70: Please add citations to this sentence

These are merely speculations/guesses that I am testing in the remainder of the manuscript. I am not making any definite statements that need to be supported by literature. That's why I preface the whole section with "The mechanism ... remains unclear".

L71: prerequisite(s)

Changed to "prerequisites" (l. 77).

L89: I think I get what you are saying here but I believe it could be worded in a more straightforward way

Changed to "microbially fixed" (l. 92).

L97: plant and macroalgal decomposition at large?

According to the definition I work with, macroalgae are plants. Please see Bolton JJ (2016) What is aquatic botany? – and why algae are plants: the importance of non-taxonomic terms for groups of organisms. Aquatic Botany 132, 1–4. <https://doi.org/10.1016/j.aquabot.2016.02.006>. I now cite this paper to avoid confusion about my use of the term "plant" (e.g. l. 70). This functional, non-taxonomic definition of "plant" greatly eases communication because it avoids constantly having to specify the inclusion of algae. When I want to be more precise or refer to taxonomic groups, I use other terms such as "vascular plants", "freshwater plants", "aquatic plants", "marine angiosperms", "brown algae" etc.

L101: diverged from seaweeds? Or from each other? And are they morphologically different from seaweeds or each other?

Seaweeds are not mentioned at the beginning of this paragraph or in most of the previous paragraph, so I am unsure where the idea that I am comparing to seaweeds is coming from. The sentence refers to "their" phylogenetic and morphological differences with "their" referring to the two beforementioned seagrass species. I believe I cannot make this clearer while remaining concise.

L107: I have not seen supplemental data presented in this way (as a citation). I would lean towards referring to it as a supplemental data file

These data will not be in the journal supplement. Only the supplementary PDF will. The data are and will be in the GitHub and Zenodo repositories. The standard way to cite these repositories is with a DOI.

L111: influence of photosynthesis?

Rephrased "physiological influence" as "influence of physiology".

L112: I believe that this is an important topic beyond just the blue carbon context. I would suggest including other implications of this research- I think that there are other physiological and ecological implications of detrital photosynthesis.

Also I do not agree that seaweeds are the "current focus" of blue carbon.

In line with the reviewer's request to minimise mention of seaweeds and blue carbon, I have decided to remove this final sentence from the aims. The rationale for the study is already laid out at the beginning of the paragraph and the sentence was somewhat repetitive. Instead I have added mention of the causal model I am presenting in Figure 4 (ll. 121ff.).

Methods

L210: Change subheading to "detrital photosynthesis model" I would also change the order of methods to match the results so first explain the seagrass exp, then the lit review, then the photosynthesis model, then the meta analysis

As mentioned above, the meta-analysis methods already exhaust the word allowance, so I moved the seagrass methods to the supplement. The meta-analysis methods consist of the model development, literature search and quantitative analysis in that order. Model development precedes seagrass data collection and the literature search for data to which the model is then fit. Since this whole study is concerned with nothing other than detrital photosynthesis, I do not understand the need to specify this in a subheading.

L121: this sentence sounds more like an introduction sentence

Yes, this sentence introduces the rationale behind previous models.

L129: Why do you say this? I do not know why we would not expect say a buoyant macroalgae detached from substrate that floats to the surface and is exposed to higher light quality + quantity to photosynthesize more optimally.

As explained above, the sentence specifies "under identical conditions" (l. 135). Everything the reviewer mentions, represents a change in conditions. I have added clarification using the reviewer's provided example: "changes in environment (e.g. when detritus floats and is exposed to more light)" (l. 136).

L166: Author needs to make one of two changes here. 1) remove the word "systematically" here (and elsewhere if it is used elsewhere) or else 2) follow the PRISMA guidelines for systematic reviews <https://www.prisma-statement.org/> Just reviewing the literature is definitely fine for the purposes of this article- but systematically reviewing the literature is a different thing. Would be ideal to get a sense of the number of papers considered for inclusion regardless (although this would be a requirement if the review is systematic).

Removed "systematically" (l. 173). The number of included papers is given in l. 190 and a complete list is in the supplement. I would like to point out that while I did not have space for inclusion of a PRISMA flowchart, my review uses explicit target and exclusion criteria and is fully reproducible because I mention the databases and keyword string that were used.

L174: define what you mean by abscission here

Abscission unambiguously means a plant actively shedding plant parts and does not need to be defined in a botany journal.

L191: so no reds? Include /Ochrophyta for browns

Correct, the text specifies that Chlorophyta and Heterokontophyta made up the seaweeds and the cited table also does not mention Rhodophyta. I did not add the alternative phylum name "Ochrophyta" because in the next sentence I mention that I follow Guiry's taxonomy (AlgaeBase) which does not use the term "Ochrophyta". Please see the paper "Validation of the phylum name Heterokontophyta" by Giury.

L205: I would also hesitate to call this a meta-analysis, maybe just data analysis? Again meta-analysis has strict requirements following PRISMA.

Meta-analysis is widely understood to mean data analysis across studies, i.e. quantitative synthesis. This is what I have done in this manuscript.

L233: why did you think that type of response variable would impact the outcome? Should you not present an a priori hypothesis prior to including it in your model? Would you consider removing this predictor to simplify the model? Also make it clear that photosynthesis as a response includes all of the different measures of photosynthesis mentioned previously.

The rationale for inclusion of these predictors is provided in the next sentences: "Light is naturally a key predictor of photosynthetic persistence (Frontier et al. 2021, 2022, Wright et al. 2024) and the type of response variable is important because chlorophyll and photosynthesis are expected to senesce on different timescales. Of the recorded predictors, water availability is only relevant to terrestrial plants and was therefore not considered..." (ll. 242–246). I generally do not think it is wise to exclude predictors when the information is available and can feed into the model (omitted variable bias). Of course, the mistaken inclusion of a second response variable as a predictor leads to post-treatment bias. I think I have made it abundantly clear how the photosynthesis response variable is calculated. The whole previous paragraph is dedicated to this, starting with "The response variable is an amalgam of various measures of photosynthesis." (l. 226) which already clarifies what the reviewer is asking me to clarify.

L237: remove sentence starting with water- many other abiotic variables were not measured/considered in your model.

Removing this sentence would obscure the fact that I did not include water as a predictor. I have clarified that water was excluded among the variables extracted from the studies (ll. 245f.).

Notes on equation presented in L245 onwards:

This section gets extremely muddled to me. I would consider if there are ways to more clearly explain your equations and variables, there are a lot of variables! Perhaps inserting text to define/describe variables between lines of equation simply in words would help. Readers would benefit from improved clarity in this section.

I have made some adjustments to remove the amount of potentially unfamiliar symbols, such as writing out the normal distributions which are familiar to everyone (ll. 257–269). There are not a lot of variables, only the ones mentioned in the text preceding Equation 2: the response variable p and the predictor variables t , G , S , E , L and C . The remainder (α , β , γ etc.) are parameters. I use standard statistical notation to describe how probability distributions relate in my model. I am now referring the reader to a textbook with that notation (ll. 254f.). To describe the model with such

accuracy in words would demand several paragraphs. I chose the journal Quantitative Plant Biology because I can expect a certain level of mathematical literacy from its readership which allows me to be appropriately succinct in the quantitative methods.

In line 245 direct readers back to previously described Equation 1 where p was defined.

I already make this link earlier, in ll. 237f.: "This normalisation allows using the simplest form of Equation 1 with $\tau = 0$ and $\alpha = 1$."

In line 246 species is included? Or is that a 5?

Capital "S" has clearly been defined to stand for species (l. 253).

L267: "Even though is suspected" typo?

Thanks for flagging. Replaced "is" with "I" (l. 279).

L285: These are reviews? So would be mean k values? What would the sample size be?

Each of the cited studies present a dataset, not always a review or meta-analysis. k values are parameters, so they typically arise from a whole series of data to which an exponential curve is fit. As mentioned above, the traditional notion of sample size breaks down here. In the model I account for variation in k values between studies and plant groups. There is no space to describe this secondary and simple statistical model with an equation like Equation 2. The description "log-normal multilevel intercept model" and code I am referring the reader to (ll. 302f.) are informative enough.

L290: How was this ratio calculated- was this calculated on a per study basis? As in only including studies that measured both photosynthesis and biomass degradation, such that each ratio is controlled for experimental/methodological/species-driven differences? Or was it one global mean value divided by one global mean value (I do not agree that this is appropriate)? Also in your results when discussing these ratios please indicate sample sizes and standard errors behind these estimates.

I do not work with means, I work with probability distributions. The ratio is therefore the ratio of two probability distributions, not two means. This approach accounts for all measurable uncertainty. I hope this is evident when looking at my figures. There are extremely few papers that report decomposition and detrital photosynthesis (less than a handful written on kelps by my colleagues on I). I am therefore independently estimating the probability distributions for the photosynthetic and detrital half-lives and formally comparing them as a ratio. I have clarified that the ratio is calculated from "posterior probability distributions" (ll. 310–313). As mentioned above, the ratio has no defined sample size because it is not a variable that can be sampled. It is a parameter. "Standard error" is a frequentist term. I do Bayesian statistics and we report the standard deviation of parameters. These can be found in the results.

Results

L297: change to seagrass photosynthetic degradation experiment

As above, since this paper is on detrital photosynthesis I think it is much more elegant just to specify that this section concerns the seagrasses within this context.

L300: based on what metric? Missing methods makes these results hard to interpret

The methods are not missing. They are in the supplement and the reader is referred to them (l. 173). As mentioned above, I am happy to move the seagrass methods to the main section at the editor's discretion. They will substantially increase the length of the manuscript.

Figure 2- I just thought of this, but wouldn't light vs dark also co-occur with field vs lab experiments? Because all dark observations likely come from the lab? Or not necessarily?

Practically all of these experiments are lab-based. I am not aware of a study completely blacking out mesh bags in the field and imagine this would be logistically difficult while keeping the detritus accessible. There are a few studies manipulating light via depth in the field. Regardless, there are not enough field studies spread across plant groups to include a field effect in the model if the reviewer is suggesting that it may confound the light effect.

Also re fig 2 I wonder if chlorophyll and photosynthesis plots could be combined? The shapes of the curves and axes are so similar and the small size of the graphs in the resulting figure make it hard to see small differences anyway

I initially plotted it this way but the uncertainty intervals are simply too wide and there are too many datapoints, so the plot becomes unreadable. I tried a large number of designs before settling on Figure 2.

Lastly for figure 2 the bottom section is not described in the caption consider making it it's own figure with proper caption.

The bottom part aligns with the plant groups (headings and colours) of the top part. As is clear from Equation 2 which I refer to in the caption, the distributions in the bottom part are parameters underlying the curves in the top part. Due to these connections, I do not think a separation into two figures would be sensible. This sentence in the caption describes the bottom section the reviewer is referring to: "Distributions are kernel density estimates of posterior probabilities for the light and chlorophyll relative effect sizes (e^{β} and e^{γ} in Equation 2)." (ll. 401ff.).

L292: Again, I do not like this phrasing "if $\mu / t^{1/2} > 1$, photosynthesis essentially determines decomposition." We cannot say this with the data presented here.

In this section of the methods I am simply describing what the ratio means, independent of the data. I have reworded in ll. 304–310.

L334: Median ratio of what?

Specified as "median light–dark ratio" (l. 353). This is the relative effect size of the light effect.

L336: what does the author mean by substantial?

This is elaborated on in the following sentence (ll. 354ff.).

L352: what does this mean "The prediction for new plants is 17 days ($\ln$ days = 2.8 ± 1.1)."

"ln" stands for natural logarithm. This is more specific than "log" which may be interpreted as $\log_{10}$. I have removed "days" to avoid confusion because logarithms are unitless (l. 369). Mean $\pm$ s.d. is reported in log space because the estimates are very right-skewed. The log-normal distribution is an example of such a right-skewed distribution. In log space it is normally distributed and can be adequately described in terms of summary statistics.

L357: this sentence belongs in discussion

Moved to discussion, as also requested by the other reviewer.

L375: add number after words. Inter group as in inter plant group?

Reworded.

L376: poorly worded sentence that belongs in discussion

Reworded and moved to discussion.

L392: where are these data coming from again? The author's review or the previously cited review papers in the methods?

The methods clearly state that data were pooled from various published datasets and analysed in a novel way. I am not quoting other reviews in my results.

L409: I just really don't like this phrasing "Terrestrial decomposition is almost certainly not influenced by physiology"- I think you should stick to talking about photosynthesis specifically, lots of other physiological processes go on during degradation, what you're looking at here is just photosynthesis not physiological functioning more broadly. And furthermore many different things impact degradation (this goes for other similar uses of the word physiology throughout the ms)

Rephrased to avoid confusion with other physiological processes mentioned by the reviewer (e.g. detritivore physiology). I am using the term "physiology" throughout to refer to plant physiology that may take place in darkness, such as induced defences, growth and repair. Please see the new Figure 4 for a more detailed picture of the causal model I am proposing.

Table 2- please include sample sizes. Please also explain why $\ln$ mean is presented? I think mean rather than median would be more appropriate, plus the corresponding standard error. Makes the table confusing for the reader to interpret to present both median and $\ln$ mean. Please describe in the first column in words what each symbol refers to.

Sample sizes are included in the first row. I have added an explanation for the use of summaries in log space (l. 381).

Discussion

L433: include these citations (a subset)

I intend this section to be a brief recap of the problem outlined in the introduction. I do not know what statement the reviewer wants me to support with citations.

L445: This is not a true. Grazing, microbial decomposition, abiotic conditions, etc all have a massive impact on seaweed degradation dynamics, we do not have enough data on degradation to say that physiology is THE main driver of degradation. This sentence needs to be modified- physiology could be one of the main drivers of degradation, along with other important drivers such as...

This is a misunderstanding about the sequence of variables on the causal path. Microbes, abiotic conditions and detrital photosynthesis are not predictors of decomposition that are independent of one another. I have created Figure 4 with a directed acyclic graph to rectify this misunderstanding and refer to this figure repeatedly (e.g. ll. 462, 501, 531).

L450: add mean photosynthesis half-life here

All relevant parameters are given in the tables. Rather than reiterating medians from the results, I prefer to make statements that add to the interpretation.

L455: If there is no evidence for this then please remove this sentence.

The fact that there is currently no empirical evidence for a mechanism does not negate its possibility.

L457: it could delay the onset of degradation.

I am not sure what the reviewer means with reference to leaf lifespan. Longer leaf lifespan may delay the conversion of a leaf to detritus, but how would this translate to faster decay of detrital photosynthesis as seen in Amphibolis?

L459: comparison is between buried vs not buried here, correct?

Correct.

L463: senescent or buried?

Correct. "between live and senescent or buried" is short for "between live and senescent or between live and buried".

L463: how is live vs dead determined here?

Buried seagrass is assumed dead. Seagrass senescence is typically assessed based on leaf colour: green leaves are live, brown or black leaves are dead.

L470: citation needed

This is one of my results.

L472: demonstrated where/by who?

Demonstrated in this study by me. I have rephrased to avoid confusion (ll. 499–505).

L473: is it the normal fate?

I think most people would agree that the leaves of land plants fall on soil rather than in water.

L474: this info should also be in the introduction

These studies are cited in the introduction but their findings are part of this review and hence only discussed here.

L475: freshwater plants from the order Alismatales

Reworded.

L476: why would this be the ultimate reason?

“Ultimate” as opposed to “proximate” in the following sentence (l. 512).

L477: Is the difference between detrital photosynthetic capacity in freshwater vs marine plants that significant? Table 3 indicates no, so why spend so much time discussing an insignificant difference? Overall isn't the main explanatory mechanism behind the ability for aquatic/marine seaweeds/plants to photosynthesize for longer the greater ability for the plants to take up nutrients in water vs. terrestrial plants are reliant on roots for water + nutrient uptake? The discussion regarding hydrogen peroxide build up seems a bit over complicated and under validated.

I do not report statistical significance. There is just more or less probability of a difference. The probability of seagrasses and freshwater plants being different in terms of photosynthetic half-life is about the same as that of freshwater and terrestrial plants being different (Table 3). As explained elsewhere, they form a spectrum. Of course, the differences are not as large as between seaweeds and the other plants, but that does not make them unworthy of discussion. More importantly, detached freshwater plant leaves are very different in their light response (Figure 2). This result deserves discussion because the mechanism of hydrogen peroxide buildup is quite well-understood.

L492: Again, this needs to be tempered with the fact that grazing and/or microbial degradation are also main drivers of macroalgal decay in situ

Please see Figure 4 and my previous explanations.

L494: *may* photosynthesize, in suitable conditions

Added “may” (l. 528).

L499: what predictions?

I restructured the last paragraphs so this is no longer mentioned.

L500: this seems like a new thought therefore new paragraph. What is the significance of green leaves anyway? Haven't you done the work to actually assess photosynthesis?

I moved mention of green seagrass leaves to the introduction (l. 98) and restructured the end of the discussion.

L509: "but many of which are not cast up on the beach to die" what do you mean by this?

L510: this is hypothetical but don't you have data that backs this up?

L515: what conventional understanding?

L520: awkward/unclear sentence

L521: Same here- needs more detail. Author is assuming that the reader knows what 'deep ocean retention' refers to in this context, and many readers may not.

This entire section has been rewritten.

L526: how would one explicitly account for detrital photosynthesis?

Theory provides a framework for understanding empirical data. If we apply the wrong assumptions, we misinterpret the data. In that sense, accounting for detrital photosynthesis may simply be knowing about it and taking it into considering when interpreting observations (ll. 547–569) or it could be an adjustment of exponential decay models as I suggest in the concluding paragraph (l. 578).

L529: I don't understand this sentence. What is the importance of tracking the fate of freshly excised tissue vs. "dead" tissue? Isn't the fresh tissue on the same trajectory as more senescent tissue?

Rephrased. No, the result is not the same. If one uses fresh tissue as a proxy for detritus, the measured decomposition rate is a lot slower.

L533: Where was this mentioned before? (Ok now I see but an entire dislodged plant does not necessarily remain viable... could still easily just decay, no? A fresh healthy kelp blade excised from a plant in a storm could be just as viable as a whole plant)

Removed.

L534: *some seaweed carbon sinks. Also you found yourself that many experiments are also done in the dark.

Rephrased in ll. 563ff. The dark experiments I included in the meta-analysis quantified photosynthesis, not decomposition. Most decomposition experiments (e.g. litterbag experiments) are carried out under ambient light conditions.

L537: I do not like this sentence. What pre-conceptions? A more specific statement ie. detrital photosynthesis ought to be incorporated into models of seaweed carbon longevity is more appropriate.

L542: And timescales vary depending on abiotic conditions

L543: Again please modify this sentence as suggested in above comments.

All rewritten.

L544: seaweeds are not plants

I and many others disagree. Research on macroalgae is routinely published in plant journals such as Nature Plants, New Phytologist, Annals of Botany, Journal of Ecology etc. See my reference above to Bolton's paper, which I did not think necessary to invoke but now repeatedly cite (ll. 70, 118, 178).

L545: Run on sentence. Author should split into 4 sentences and provide more details. How would one measure photosynthesis in situ? What is the importance of studying the "initial physiological state" of detritus and what does "initial physiological state" really mean? For 2) how does this advance our understanding of carbon dynamics? Should we be striving for one single model of macroalgal degradation when it is so variable across contexts?

I am presenting a list of future research objectives, so they are numbered in a single sentence on purpose. I elaborated on the "initial physiological state" (ll. 548ff.). It means the level of photosynthesis displayed by detritus when it is produced, i.e. at the onset of the detrital phase at detachment. No, a single model species would be misleading because even differences within genera are large (see Wright LS, Pessarrodona A, Foggo A (2022) Climate-driven shifts in kelp forest composition reduce carbon sequestration potential. Global Change Biology 28, 5514–5531. <https://doi.org/10.1111/gcb.16299>).

Supplement

What timepoints (day numbers) were sampled?

This can be gleaned from Figure 1 and the data which are openly available on GitHub. The exact sampling dates varied between species and is irrelevant to the model. All that is relevant is the duration and number of timepoints which are mentioned.

Add citations to table S3 please

Table S3 presents parameter estimates for species across studies. There are therefore no unique citations for each row. Please see the openly available dataset (Zenodo and GitHub) which cites each observation extracted from the literature with a DOI and citation.

What is Exper. vs Observ. in Table S1? Please have header repeat on each new page for multi page tables. Maybe you could use color coding to indicate marine vs. freshwater vs. terrestrial vs. seaweed here?

The table header specifies "The longest experimental durations and largest numbers of studies, experiments and observations in each non-taxonomic group are highlighted in bold.", referring to experiments and observations. "exper." and "observ." are the standard Oxford English dictionary abbreviations for these words (see <https://www.oed.com/information/understanding-entries/abbreviations/>). I have reduced the margins to 2 cm to give the table more space, but using standard abbreviations remains the best solution. I now repeat the header row on each page for

multi-page tables. I prefer not to use colour in tables because black text is easiest to read and coloured cell backgrounds may obscure text. Besides, only a single distinction needs to be made here between freshwater and marine Alismatales which are a minor section of the table, so a single symbol such as the asterisk suffices.

Reviewer: 2

Comments to the Author

The manuscript entitled “Plants in Limbo – The Theory of Detrital Photosynthesis” addresses an interesting and timely topic and is, overall, well written. The Introduction clearly outlines the conceptual background and effectively identifies the existing knowledge gaps. The Methods section is described in sufficient detail, and the provision of database access, methodological documentation, and R code substantially enhances transparency and reproducibility.

Thank you.

The Results are presented clearly; however, parts of this section contain interpretative elements and narrative explanations that would be more appropriately placed in the Discussion. Revising these passages to maintain a strictly results-focused structure would improve clarity and organization. The figures are generally well prepared and informative. Nevertheless, the presence of empty panels in Figure 2 requires clarification. The rationale for these empty panels should be explicitly stated, either in the figure caption or in the Methods section.

Removed all interpretative sentences to discussion. Empty panels are now explained in methods (ll. 246ff.) and figure caption (ll. 400f.).

The Discussion is coherent and well developed. However, it would benefit from a more explicit comparison between decomposition and photosynthetic processes in plant species that are evolutionarily adapted to this mode of existence, particularly submerged pleustophytes. Framing such a comparison within the theoretical context presented in the manuscript could provide additional mechanistic insight and strengthen the conceptual contribution of the study. Although this suggestion is exploratory, and acknowledging that representative taxa (e.g., from the genus *Ceratophyllum*) are primarily freshwater, this perspective may nonetheless offer a valuable extension to the proposed framework.

The discussion has been rewritten. I thank the reviewer for their interesting insight into pleustophytes. Pleustophytes are now mentioned in the introduction and methods (ll. 51, 183), but I cannot see how they can be incorporated into this theoretical framework because the definition of detritus as detached plant material does not apply to them. I cannot imagine how the framework would be extended to include pleustophytes because detrital age which starts with experimental excision or detachment in nature is undefined for these plants (l. 184). I suppose they just reach the end of their lifespan and then die and sink, so the senescence process cannot be split into two distinct phases: attached and detached senescence.

The detailed comments and suggestions:

Graphical abstract:

The graph, in its current form, is not sufficiently informative. It is unclear what exactly is being presented and how the displayed elements convey the intended information. In my view, if this figure is

essential to the manuscript, it should present the actual data values rather than relying primarily on shaded boundaries or ranges, which may obscure interpretation.

I have added black lines to denote medians and guide the viewer's eye. I do want to retain the whole probability distribution around each median. Just showing the medians would be misleading. This figure is not essential to the manuscript. It visualises data presented in Figure 3 in a simplified way to approximately answer the two questions posed in its title. The main function of the graphical abstract is not to convey complex information but to provide a simple visual summary and engage the reader.

Line 53: What does it mean? The term exists, so the baseline is related to what? Please specify.

Changed to "reference point" (l. 56), meaning among other plant groups than seaweeds. This is a topic statement which is elaborated in the remainder of the paragraph.

Line 121-122: Please specify what baseline means here.

Reworded (ll. 127f.).

Line 160-163: Interesting explanation of the parameter symbols chosen.

Thank you. I am glad this is appreciated.

Line 216: The space is not needed.

Changed font and spacing. I was hoping this code could be parsed as code by the copyeditors, hence I had highlighted it with different font and spacing.

Line 309-313: This is rather a discussion sentence than a results section and should be moved to the discussion section.

I am describing my observations of periphyton here, which is a result. I changed the sentence order to lead with the periphyton observation, but it is unsensible not to provide context for the reader in the subsequent sentence. They might otherwise wonder why I bother to mention periphyton.

Line 334-335: Why is the light photosynthesis panel of freshwater plants in Fig. 2 empty? This needs to be explained either in the main text or in the Fig. 2 caption.

Explained in methods (ll. 246ff.) and figure caption (ll. 400f.).

Line 358-363: This is rather a discussion sentence than a results section and should be moved to the discussion section.

Moved to discussion, as also requested by the other reviewer.

Line 377-379: This is rather a discussion sentence than a results section and should be moved to the discussion section.

Reworded and moved to discussion.

Figure 2 Why are Freshwater Light and Seagrasses Dar panels empty? This needs an explanation.

Line 392-394: This is rather a discussion sentence than a results section and should be moved to the discussion section.

Reworded (l. 405).

Line 410-413: This is rather a discussion sentence than a results section and should be moved to the discussion section.

Moved to discussion.

Line 431: The discussion would benefit from a more explicit comparison between decomposition and photosynthetic processes in plant species that are evolutionarily adapted to this mode of existence, particularly submerged pleustophytes. Incorporating such a comparison, set against the theoretical framework presented and discussed in the manuscript, could provide additional insight into the mechanisms under consideration. Although this suggestion is exploratory, and recognizing that representative taxa (e.g., from the genus *Ceratophyllum*) are primarily freshwater, such a perspective may still offer a useful conceptual extension.

I thank the reviewer for their suggestion. I now mention pleustophytes alongside holopelagic species in the introduction (l. 51) and methods (l. 183). I explicitly exclude them because the term "detritus" is not defined for these species except in the strict sense as dead organic matter. They are always detached, so the common definition of detritus as detached plant tissue does not work. One would have to define the onset of the detrital phase with some other criterion, such as (1) time of sinking (some papers I cite measure this), (2) time of death (detritus sensu stricto) or (3) a case-specific definition such as stranded wrack. It is therefore not possible to define detrital photosynthesis as separate from ordinary photosynthesis for pleustophytes and holopelagic algae.

Sincerely,
Luka Seamus Wright